AgenticAI Foundry

Student Quick Start Guide

MIT Professional Education | Applied Generative AI for Digital Transformation

This guide gets you from zero to running the course demos in the shortest possible time.

What's Included

Demo	Module	What You Will Learn	API Key?
LLM Cost Explorer	Module 1	Why the same AI task can cost $1 or $230 depending on model choice	No
Multi-Agent Demo	Module 2	How three AI agents collaborate like a team (CrewAI)	Optional
LangChain Agent Demo	Module 2	How a single agent uses tools to answer questions in real time	Optional
MCP Explorer	Module 3	How AI agents connect to external tools (calendars, CRMs, etc.)	No
Agent Security Demo	Module 4	Prompt injection attacks and defense-in-depth guardrails	Demo: No / Live: Optional
HITL Demo	Module 5	Patterns for keeping humans in control of AI decisions	No

Modules 1, 3, 4, and 5 work immediately - no account or API key needed. Module 2 needs Ollama (free, local) or an OpenAI API key.

Step 0 - Get the Code (Both Paths)

You need a copy of this repository on your computer before doing anything else.

Option 1: Download ZIP (easiest)	Option 2: Clone with Git
1. Click the green <> Code button on GitHub 2. Click Download ZIP 3. Find it in Downloads and extract it	`git clone https://github.com/dlwhyte/AgenticAI_foundry.git`

Which path? "Have I used Docker before, or am I comfortable installing a new app?"

Yes - Path A: Docker (recommended) No - Path B: Python

Path A: Docker (Recommended)

Docker packages the entire app into a self-contained box. Runs identically on Windows, Mac, and Linux. Time: about 20 minutes first time, under 1 minute after that.

Step	Action	Command / Detail
1	Install Docker Desktop	Windows/Mac: docker.com/products/docker-desktop Linux: see docs/DOCKER_GUIDE.md
2	Open a Terminal	Windows: Win+R, type powershell Mac: Cmd+Space, type Terminal Linux: Ctrl+Alt+T
3	Navigate to folder	cd path/to/AgenticAI_foundry (Tip: drag the folder onto the terminal)
4	Build the app (once)	`docker build -t agenticai-foundry .`
5	Run the app	`docker run -p 8501:8501 agenticai-foundry`
6	Open in browser	http://localhost:8501
7	Stop the app	Press Ctrl+C in the terminal

Docker Troubleshooting

Problem	Solution
"Docker command not found"	Make sure Docker Desktop is open and running
"Cannot connect to daemon"	Open Docker Desktop; wait for the whale to stop animating
"Port 8501 already in use"	Run: docker run -p 8502:8501 agenticai-foundry then open :8502
Build seems stuck	First build downloads ~500MB - can take 5+ minutes on slow connections

For more detail see docs/DOCKER_GUIDE.md in the repo.

Path B: Python

Run the app directly. More steps but full code visibility. Time: ~15 minutes.

Check Python first: run `python3 --version` in a terminal. Python 3.9 or lower? Use Docker instead. CrewAI (Module 2) needs Python 3.10+.

Step	Action	Detail
1	Install Python 3.10+	Check version: python3 --version If 3.9 or lower: download from python.org/downloads Windows: check "Add Python to PATH" during install
2	Open terminal and navigate	See Docker Path Steps 2-3 for opening a terminal and using cd to reach the folder

3	Install dependencies (REQUIRED - both)	`pip3 install -r requirements.txt`
		`pip3 install -r requirements-crewai.txt`
		Run BOTH commands. Takes 2-5 minutes each.

4	Run the app	`python3 -m streamlit run Home.py` Browser opens to http://localhost:8501 automatically
5	Stop the app	Press Ctrl+C

Important: Step 3 runs TWO pip commands, not one. requirements-crewai.txt contains CrewAI, LangChain, and agent dependencies. Skipping it causes Module 2 to crash with ModuleNotFoundError.

Python Troubleshooting

Problem	Solution
"pip not found"	Try pip3 install -r requirements.txt instead
"streamlit not found"	Try python -m streamlit run Home.py
"Permission denied"	Add --user: pip install --user -r requirements.txt
Browser does not open	Manually go to http://localhost:8501
Module 2 crashes on import	Make sure you ran BOTH pip install commands in Step 3

Setting Up the Agent Demos (Module 2)

Modules 1, 3, 4, and 5 work immediately. Module 2 needs an AI model - you have two options:

Option A: Ollama - Free, runs locally, no account needed

Step	Command
1. Install Ollama	Download from ollama.ai
2. Download model	ollama pull llama3.2 (2GB one-time download)
3. Start Ollama	ollama serve (keep this terminal open)
4. Run the app	Use Docker or Python path, then select Ollama in the app sidebar

The model download is ~2GB. Do this on a good internet connection.

Option B: OpenAI - Paid, faster results (~$0.01 per demo run)

Create an account at platform.openai.com and get an API key. Enter it in the app sidebar - no environment setup needed.

What to Expect When It's Working

Open http://localhost:8501 to see the AgenticAI Foundry home with all 6 modules. Navigate via the left sidebar. Run python setup_check.py to verify your environment.

Module	What You Should See
Module 1: LLM Cost Explorer	Interactive cost comparison charts - no API key needed
Module 2: Multi-Agent Demo	Requires Ollama running or OpenAI key in sidebar
Module 2: LangChain Agent Demo	Live crypto price lookups - requires Ollama or OpenAI
Module 3: MCP Explorer	Interactive MCP protocol diagram - no API key needed
Module 4: Agent Security Demo	Prompt injection demonstrations - no API key for demo mode
Module 5: HITL Demo	Human-in-the-Loop approval patterns - no API key needed

Documentation and Getting Help

Guide	What It Covers
Student Quick Start (this PDF)	Screenshots walkthrough - start here if you're new
docs/DOCKER_GUIDE.md	Full Docker setup with troubleshooting
docs/BEGINNERS_GUIDE.md	Deep explanation of all technologies used
docs/CREWAI_SETUP.md	Ollama and OpenAI setup for agent demos (Module 2)
docs/HITL_GUIDE.md	Human-in-the-Loop patterns (Module 5)
docs/LLM_COST_GUIDE.md	Token economics and model pricing (Module 1)
docs/MULTI_AGENT_GUIDE.md	CrewAI vs LangChain patterns (Module 2)
docs/MCP_GUIDE.md	Model Context Protocol explained (Module 3)

Getting Help

- Setup issues: Run python setup_check.py and share the output
- Docker problems: See docs/DOCKER_GUIDE.md troubleshooting section
- Agent demo issues: See docs/CREWAI_SETUP.md troubleshooting section
- General questions: Post in the course discussion forum with any error messages

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Check Your Environment

Not sure if everything is set up? Run from the project folder:

```
python setup_check.py
```

Checks Python, libraries, Docker, Ollama, and API keys - reports in plain English.